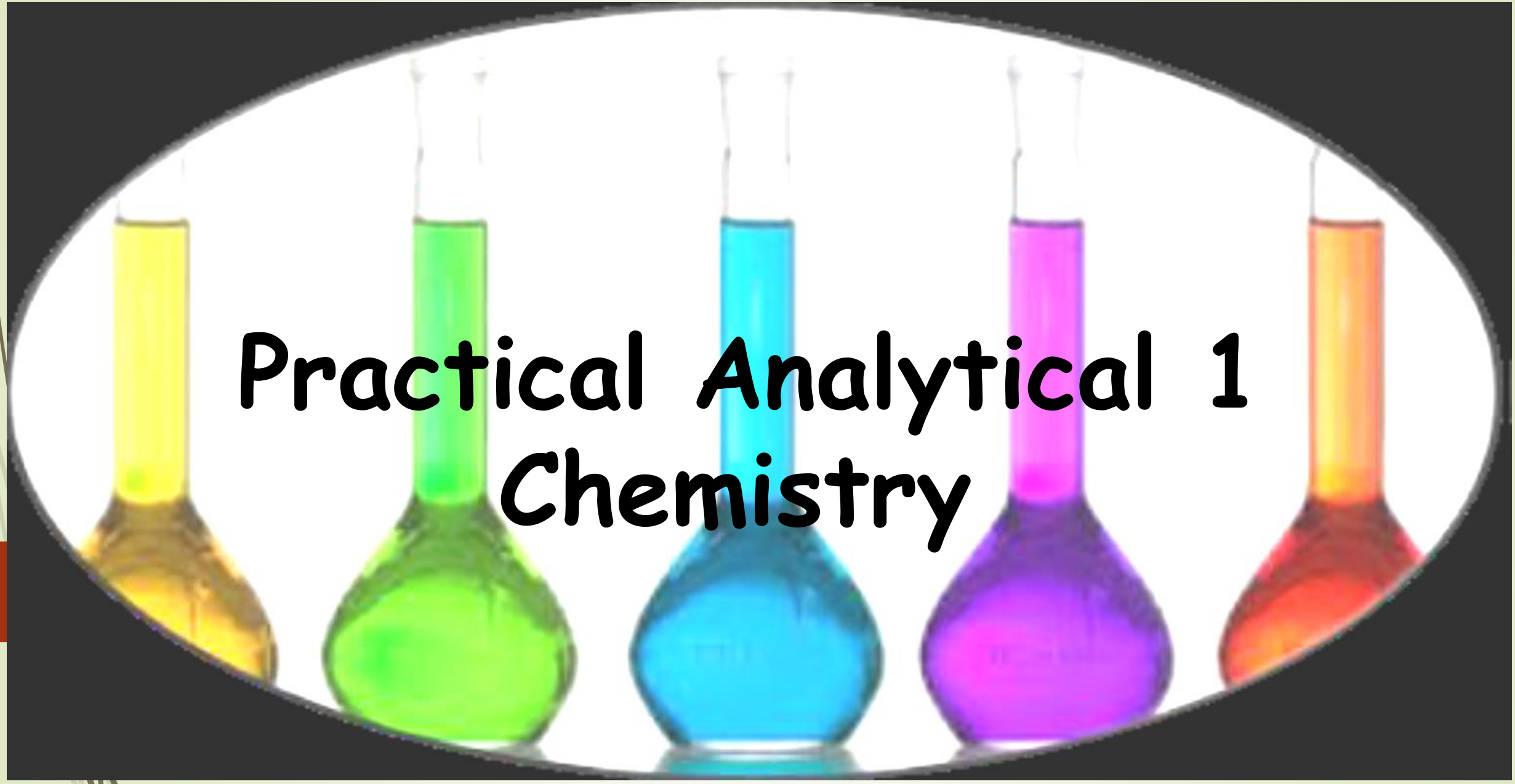


Practical Analytical 1 Chemistry

Experimental 3:

Standardization of H_2SO_4 using standard solution of NaOH

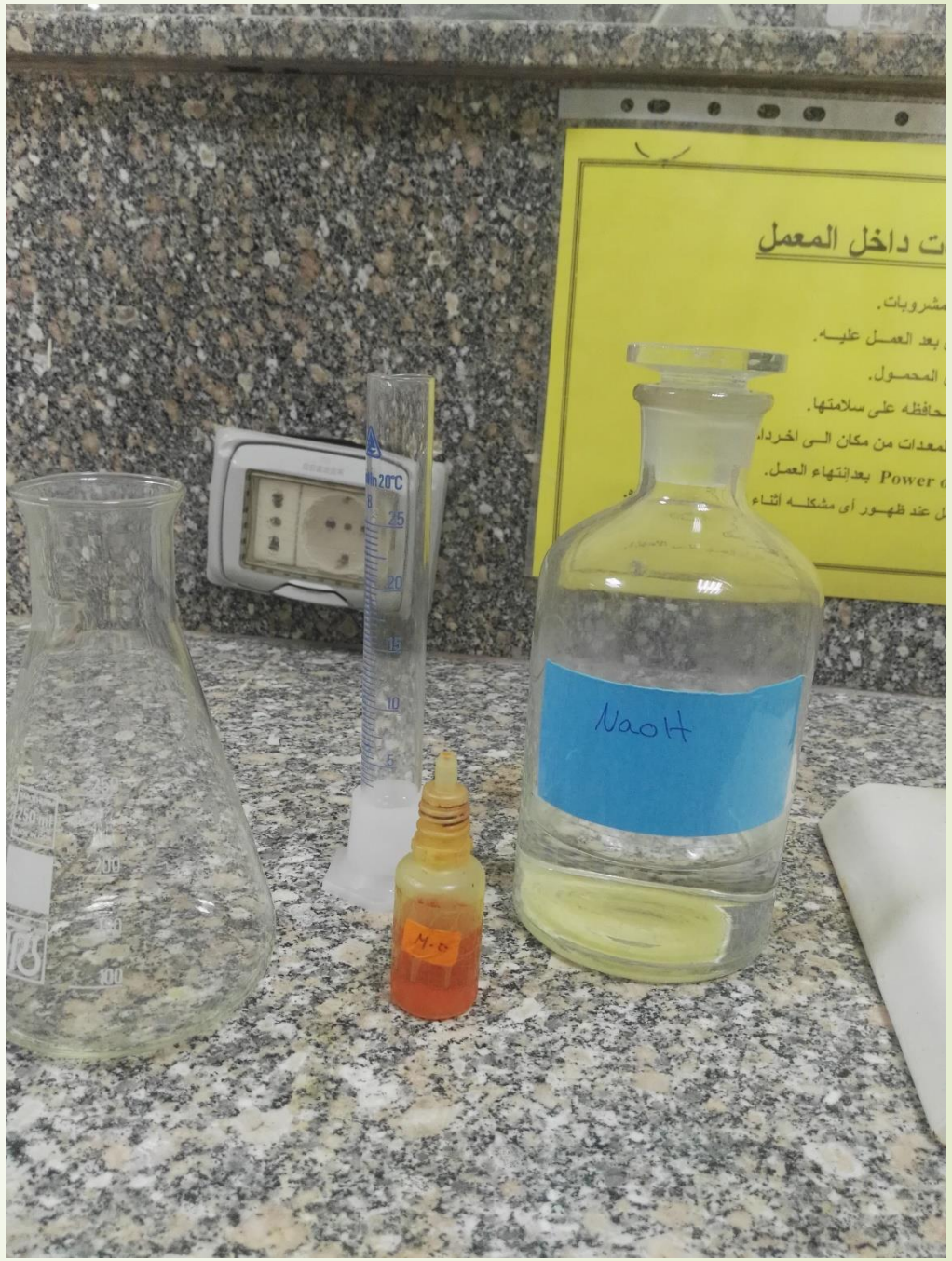
Chemicals and tools:

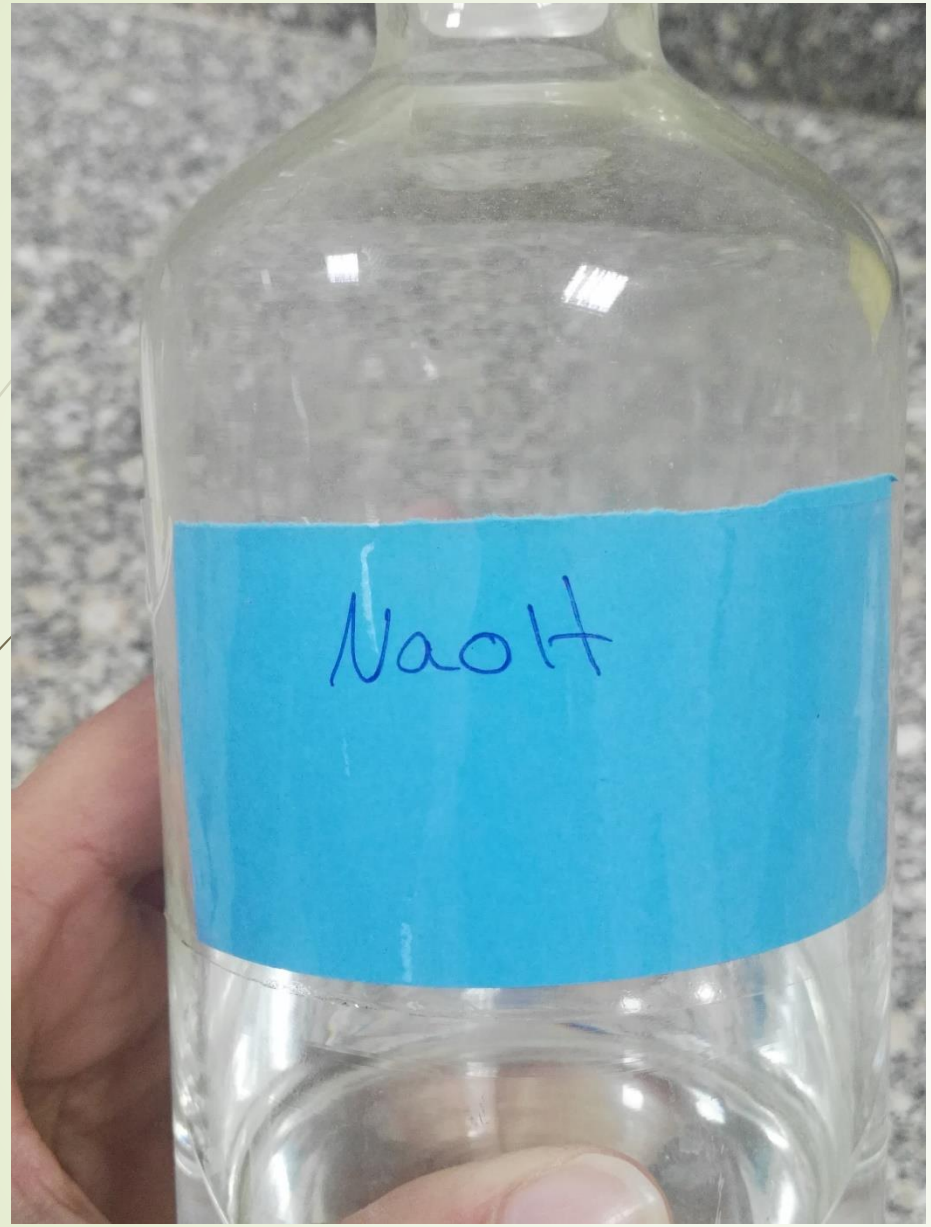
- 25-mL Burette
- 250-mL conical flasks
- Burette funnel
- 10-mL pipette
- Graduated Cylinder
- $H_2SO_4(aq)$
- $NaOH(aq)$
- Methyl Orange (M.O.)
- Distilled water

Procedure:

- Add the acid solution to the burette.
- Draw 10 mL of the base solution into the volumetric pipette and transfer this solution into a conical flask.
- Add 1-2 drops of M.O to the base solution in the flask.
- Place the flask under the burette and start adding the acid solution to the conical flask, until the color change from yellow to orange.
- Record the final reading of the burette. Wash the contents of the flask down the drain with water.
- Calculate the concentration of NaOH.

H2SO4


NaOH


NaOH

M.O

Calculation

- Suppose that the volume of H_2SO_4 is V1 and its normality is N1 while V2 is the volume taken from NaOH and N2 is its normality.
- The volume of H_2SO_4 (from burette) = V1
- $(N1 \times V1)_{H2SO4} = (N2 \times V\ 2)_{NaOH}$
- $(0.1 \times V1_{exp})_{H2SO4} = (N2 \times 10)_{NaOH}$
- $0.1 \times 10.5 = N2_{NaOH} \times 10$
- $N2_{NaOH} = 0.1\ N$
- Concentration in gm/L = normality x eq. wt.
- **Concentration of NaOH = N1 x 40= 0.1 x 40= 4.0 g/L**

Problems

Choose the right answer:

1. Equivalent weight for $(NH_4)_2CO_3$ is it 96 divided on 2
   (true or false)

2. To calculate ppm for (0.3 gm/ml) of H_3PO_4 it will be 300000 (gm/10^6 ml)
   (true or false)

3. The colors of CH_3COONH_4 in M.O. and Ph.Ph is yellow and colorless.
   (true or false)